When Does Observational Coherence Determine Probability?

Patrick S. Mahon

Abstract

When does observationally coherent data determine a genuine probability measure? We work with a directed system of Boolean algebras carrying a compatible family of finitely additive charges. Coherence and finite additivity are purely finitary conditions; the question is what further condition licenses the passage to σ-additivity.

Finitary coherence alone does not suffice: the finite–cofinite charge on $\mathbb{N}$ satisfies every structural condition yet assigns persistent mass to a sequence of events that vanishes. More generally, σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12]. Some condition beyond finitary coherence is therefore genuinely required.

We call this condition *collective exhaustion* (CE). The main result is a four-way equivalence: CE, σ-additivity of each marginal, vanishing of the purely finitely additive part in the Yosida–Hewitt decomposition, and concentration of the Stone–space measure on realised states are all equivalent. Two constructions — Carathéodory and Stone — realise the extension and yield the same measure.

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1 Introduction

A probability measure on a field $\mathcal{F} \subseteq 2^\Omega$ is a set function $P : \mathcal{F} \to [0,1]$ satisfying [1, 10]:

$$\text{(positivity)} \qquad P(A) \geq 0; \tag{1}$$

$$\text{(normalisation)} \qquad P(\Omega) = 1; \tag{2}$$

$$\text{(countable additivity)} \qquad P\Big(\bigcup_{n=1}^\infty A_n\Big) = \sum_{n=1}^\infty P(A_n) \quad (\text{disjoint } A_n,\ \bigcup A_n \in \mathcal{F}). \tag{3}$$

For a finitely additive P, (3) is equivalent to (e.g. [1], Theorem 2.1)

$$E_n \downarrow \varnothing \quad \Longrightarrow \quad P(E_n) \to 0 : \tag{4}$$

mass cannot persist along a vanishing sequence of events. Kolmogorov's extension theorem [10] and Carathéodory's [4] both take σ-additive marginals as input, bundling (3) into the hypothesis.

The directed structure is already present in the classical problem: compatible finite-dimensional distributions live on a directed family of coordinate projections. We isolate that structure — a directed system $(\iota, \leq)$ of event algebras $\mathcal{E}_i$ with surjective refinement maps

$$\pi_{ij} : O_j \to O_i \qquad (i \leq j), \qquad \pi_{ik} = \pi_{ij} \circ \pi_{jk} \qquad (i \leq j \leq k)$$

— and ask what the directed structure alone determines. Positivity and normalisation become local: each ℓ_i is a normalised finitely additive charge on $\mathcal{E}_i$. Compatibility replaces agreement of marginals:

$$\ell_j\big(\pi_{ij}^{-1}(A)\big) = \ell_i(A) \qquad (i \leq j,\ A \in \mathcal{E}_i). \tag{5}$$

These are finitary. Countable additivity (3) is not.

The finite–cofinite algebra on $\mathbb{N}$ shows how it fails. Set $\mu(A) = 0$ for finite A, $\mu(A) = 1$ for cofinite A. Then μ satisfies (1)–(5), yet

$$E_n = \{n, n{+}1, \dots\} \downarrow \varnothing, \qquad \mu(E_n) = 1 \;\; \forall\, n:$$

mass persists because $\bigcap_n E_n = \varnothing$ is a countable fact no finite check can see. In general, σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12, 8, 5]. If σ-additivity is to be *recovered* rather than assumed, the condition must fall on the charges themselves.

We call this condition *collective exhaustion* (CE):

$$E_n \in \mathcal{E}_i, \quad E_n \downarrow \varnothing \quad \Longrightarrow \quad \exists\, j \geq i \text{ s.t. } \ell_j\big(\pi_{ij}^{-1}(E_n)\big) \to 0. \tag{6}$$

CE is the exact admissibility criterion: a compatible charge family extends to a unique σ-additive measure on the observable σ-algebra if and only if it satisfies (6) (Theorem 6.1). Two constructions realise the extension — one by Carathéodory (§3), one by Stone duality (§5).

2 Observational structure and compatible charges

Definition 2.1 (Observational structure)**.** An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathsf{O}_i, \mathcal{O}_i)$, the *outcome space at level i*;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Example 2.2 (Successive observations of a dynamical system)**.** Let $T : X \to X$ and $h : X \to S$ with S finite. Set $\iota = (\mathbb{N}, \leq)$,

$$\mathsf{O}_L = S^L, \qquad \mathrm{eval}_L(x) = \big(h(x),\, h(Tx),\, \dots,\, h(T^{L-1}x)\big),$$

and let $\pi_{ML} : S^L \to S^M$ drop the last $L - M$ entries. Passing from level L to $L + 1$ appends one observation:

$$\mathrm{eval}_{L+1}(x) = \big(\mathrm{eval}_L(x),\, h(T^L x)\big).$$

The canonical instantiation is $\Omega = S^{\mathbb{N}}$ and a compatible family $\{\ell_L\}$ is a consistent system of finite-dimensional distributions of $h, h \circ T, h \circ T^2, \dots$

Definition 2.3 (Instantiation; query system)**.** An *instantiation* is a set Ω with *evaluation maps* $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ satisfying

$$\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i \qquad (i \leq j).$$

A *query system* is an observational structure together with an instantiation.

Every observational structure admits a canonical instantiation: the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathsf{O}_i = \Big\{ \omega \in \prod_{i \in \iota} \mathsf{O}_i \;\Big|\; \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j \Big\},$$

with $\mathrm{eval}_i(\omega) = \omega_i$. In Example 2.2, $\Omega = S^{\mathbb{N}}$ is the space of complete observed trajectories: each $\omega \in \Omega$ is a full record

$$\omega = \big(h(x),\, h(Tx),\, h(T^2x),\, \dots\big)$$

and $\mathrm{eval}_L(\omega) = (\omega_1, \dots, \omega_L)$ extracts the first L entries.

Definition 2.4 (Common coarsenings)**.** The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$.

Definition 2.5 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{O}_i\}$, and the *observable σ-algebra* is $\sigma(\mathcal{C})$.

In Example 2.2, $\mathrm{Cyl}(L, A)$ is the set of trajectories whose first L observations fall in $A \subseteq S^L$:

$$\mathrm{Cyl}(L, A) = \big\{\omega \in S^{\mathbb{N}} : (\omega_1, \ldots, \omega_L) \in A\big\}.$$

Lemma 2.6 (Cylinder π-system)**.** *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. The coherence condition gives $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \leq j$: the same subset of Ω is a cylinder at any finer level. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. $\square$

For each $i \in \iota$, let B_i denote the Boolean algebra of level-i cylinders $\{\mathrm{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i \in \iota} B_i$. For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

Definition 2.7 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i \in \iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \leq j \text{ and all } E \in B_i.$$

In Example 2.2, compatibility says that the probability assigned to an event described by the first M observations does not change when the window is extended to $L > M$:

$$\ell_L\big(\{s \in S^L : (s_1, \ldots, s_M) \in A\}\big) = \ell_M(A) \qquad (A \subseteq S^M).$$

Compatibility ensures a well-defined charge on $\mathcal{C}$ via $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$.

Definition 2.8 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ is surjective for every $i \in \iota$.

Lemma 2.9 (Surjectivity and injectivity from Surjective Evaluation)**.** *If the query system satisfies Surjective Evaluation, then for every $i \leq j$:*

(i) the refinement map $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ is surjective;

(ii) the connecting map $\varphi_{ij} : B_i \to B_j$, defined by $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$, is an injective Boolean algebra homomorphism.

Proof. For (i): given $y \in \mathsf{O}_i$, surjectivity of eval_i yields $\omega \in \Omega$ with $\mathrm{eval}_i(\omega) = y$. Then $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega) = y$ by the coherence condition, so π_{ij} is surjective.

For (ii): φ_{ij} is a Boolean homomorphism by construction. If $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \varphi_{ij}(\mathrm{Cyl}(i, A'))$, then $\pi_{ij}^{-1}(A) = \pi_{ij}^{-1}(A')$ in O_j. Applying π_{ij} and using surjectivity gives $A = A'$, so φ_{ij} is injective. $\square$

3 Direct extension from compatible probability marginals

Kolmogorov's extension theorem [10] takes as input compatible σ-additive marginals

$$(\pi_{F'F})_\# \nu_{F'} = \nu_F \qquad (F \subseteq F' \subset T,\ |F'| < \infty), \qquad \nu_F \in \mathcal{P}(\mathbb{R}^F),$$

and extends to a unique measure on $\mathbb{R}^T$ via tightness. Compatibility (5) and σ-additivity of each ν_F are bundled into the hypothesis. Retaining both, what replaces tightness? An order-theoretic condition on ι:

Definition 3.1 (Sequential common refinements)**.** The index set $(\iota, \leq)$ admits *sequential common refinements* if every countable $\{i_n\}_{n\geq 1} \subset \iota$ has an upper bound in ι.

Any measure that agrees on cylinders is determined on $\sigma(\mathcal{C})$:

Lemma 3.2 (Observational determination)**.** *If $(\iota, \leq)$ admits common coarsenings and P, P' are probability measures on $(\Omega, \sigma(\mathcal{C}))$ with $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all i, then $P = P'$.*

Proof. P and P' agree on $\mathcal{C}$; apply the π–λ theorem to $\sigma(\mathcal{C})$. $\square$

Theorem 3.3 (Carathéodory observational extension)**.** *Suppose $(\iota, \leq)$ admits common coarsenings and sequential common refinements, the query system satisfies Surjective Evaluation, and $\{\nu_i\}$ is a compatible family of probability measures on $\{(\mathsf{O}_i, \mathcal{O}_i)\}$. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ with*

$$(\mathrm{eval}_i)_\# P = \nu_i \qquad \textit{for every } i \in \iota.$$

Proof. Content. The assignment $\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A)$ is well-defined by compatibility: if $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$, pass to a common upper bound $k \geq i, j$ and use

$$\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B).$$

Finite additivity of each ν_k gives finite additivity of μ_0.

σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(i_n, A_n)$. Sequential common refinements yield a single k with $k \geq i$ and $k \geq i_n$ for all n. Compressing to level k:

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k\big(\pi_{ik}^{-1}(B)\big) \leq \sum_{n=1}^{\infty} \nu_k\big(\pi_{i_n k}^{-1}(A_n)\big) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Extension. The cylinder family $\mathcal{C}$ is a π-system (Lemma 2.6) closed under relative differences — for $E = \mathrm{Cyl}(i, A)$ and $F = \mathrm{Cyl}(j, B)$,

$$E \setminus F = \mathrm{Cyl}\big(k,\ \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)\big) \in \mathcal{C}$$

— so $\mathcal{C}$ is a semiring. Carathéodory's theorem (Bogachev [2], I.1.11.9) extends μ_0 to a measure P on $\sigma(\mathcal{C})$.

Uniqueness. Any two extensions agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by the π–λ theorem. $\square$

4 Collective Exhaustion

Theorem 3.3 assumes σ-additive marginals. What if only finitely additive charges are given? At a single algebra, a finitely additive charge extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if it satisfies (4) (Halmos [7]). But verifying $\bigcap_n E_n = \varnothing$ requires access to $\sigma(\mathcal{E})$, which $\mathcal{E}$ alone cannot see. In a directed system, a finer level might witness what a coarser level cannot — but the following example shows it need not:

Example 4.1 (Finite–cofinite obstruction)**.** On $\mathbb{N}$, set $\mu_0(A) = 0$ for finite A, $\mu_0(A) = 1$ for cofinite A. Then μ_0 is normalised and finitely additive, but $E_n = \{n, n{+}1, \ldots\} \downarrow \varnothing$ with $\mu_0(E_n) = 1$ for all n.

The condition that rules out this failure is:

Definition 4.2 (Collectively exhaustive family)**.** The family is *collectively exhaustive* if for every $i \in \mathcal{I}$ and every $E_n \downarrow \varnothing$ in $\mathcal{E}_i$,

$$\exists\, j \geq i \text{ such that } \ell_j\big(\pi_{ij}^{-1}(E_n)\big) \to 0.$$

No first-order condition can force CE: σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12, 11]. It is, however, the exact condition needed:

Theorem 4.3 (CE Characterisation)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every i.

Proof. (i) $\Rightarrow$ (ii). Fix i and $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. Compatibility gives $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$, so ℓ_i satisfies (4) and extends.

(ii) $\Rightarrow$ (i). If ℓ_i extends to μ_i, then $\ell_i(E_n) = \mu_i(E_n) \to 0$ by σ-additivity. Take $j = i$. $\square$

The Yosida–Hewitt decomposition [15] makes the content precise:

$$\ell = \ell_c + \ell_p, \qquad \ell_c \text{ countably additive}, \quad \ell_p \text{ purely finitely additive}.$$

CE is $\ell_p = 0$.

5 Stone realization from finitely additive data

The Carathéodory construction assumes σ-additive marginals. Starting from finitely additive charges alone, one completes instead:

$$\{\ell_i\} \xrightarrow{\text{Stone}} \hat{\mu} \text{ on } \mathrm{St}(\mathcal{C}) \xrightarrow{\text{CE}} P \text{ on } (\Omega,\, \sigma(\mathcal{C})).$$

Compactness of $\mathrm{St}(\mathcal{C})$ supplies σ-additivity at the first arrow. CE controls the second: the completed space contains ideal points that no $\omega \in \Omega$ realises, and CE is the condition ensuring the measure concentrates on the realised part.

Proposition 5.1 (Stone duality for the direct limit)**.** *The Stone space of the direct limit satisfies* $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.

Proof. By Lemma 2.9(ii), the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [9, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. $\square$

Proposition 5.2 (Inverse limit measure)**.** *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [3], §6, or [7], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [4, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. ☐

Definition 5.3 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Write $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ for the principal ultrafilter at ω, and $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the basic clopen. Discriminability makes pure injective.

Proposition 5.4 (Structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [9, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability (Definition 5.3) implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. ☐

We work with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$ throughout; Choksi's theorem applies to Baire measures on compact spaces.

Proposition 5.5 (Support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is supported on $\mathrm{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [15, 13], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. By Theorem 4.3, CE is equivalent to $\mu_p = 0$, i.e. μ is itself σ-additive on $\mathcal{C}$.

A σ-additive probability on a Boolean algebra $\mathcal{C}$ extends uniquely to a measure on $\sigma(\mathcal{C})$. Under the Stone correspondence, this measure lifts to a Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$. For any non-principal ultrafilter $u \in \mathrm{St}(\mathcal{C}) \setminus \mathrm{pure}(\Omega)$, every clopen neighbourhood $[E]$ of u satisfies $E \neq \varnothing$ in $\mathcal{C}$; but because u is non-principal, for each $\omega \in \Omega$ there exists $E_\omega \in \mathcal{C}$ with $\omega \in E_\omega$ and $E_\omega \notin u$. In particular, u is not an evaluation ultrafilter. Discriminability ensures that the principal ultrafilters separate elements of $\mathcal{C}$, so $\mathrm{pure}(\Omega)$ is dense in $\mathrm{St}(\mathcal{C})$; but the support of a σ-additive measure need not concentrate on the dense set without further argument.

The required link is classical [13], Ch. 10: for a σ-additive charge μ on a Boolean algebra $\mathcal{C}$ realised as a field of subsets of Ω, the corresponding Baire measure on $\mathrm{St}(\mathcal{C})$ assigns full mass to those ultrafilters that are closed under countable intersection in $\mathcal{C}$. Each principal ultrafilter $\mathrm{pure}(\omega)$ is closed under countable intersection (it evaluates at a point of Ω), so $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. $\square$

Theorem 5.6 (Stone observational extension)**.** *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{O}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\text{Stone}} \{\hat{\mu}_i\} \xrightarrow{\text{Prop. 5.2}} \hat{\mu} \text{ on } \mathrm{St}(\mathcal{C}) \xrightarrow{\text{Prop. 5.5}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.1, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.2, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.5, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since pure is injective (discriminability), define the descended measure P on $\sigma(\mathcal{C})$ by

$$P(A) \;=\; \hat{\mu}\big(\mathrm{pure}(A)\big) \qquad \text{for } A \in \sigma(\mathcal{C}).$$

By Proposition 5.4, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$; equivalently, pure maps $\sigma(\mathcal{C})$-sets into Baire sets, so P is well-defined. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Proposition 3.2, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to surjective evaluation, discriminability, and CE. $\square$

6 Synthesis

Kolmogorov's hypothesis — compatible σ-additive marginals — decomposes into a finitary part (compatibility (5)) and an infinitary part (countable additivity (3)). The three preceding sections show these have equivalent formulations at each layer of the construction:

Theorem 6.1 (Main equivalence)**.** *Let $\{\ell_i\}$ be a compatible family of normalised finitely additive charges on a directed system satisfying the hypotheses of Theorems 3.3 and 5.6. The following are equivalent:*

(i) (Kolmogorov) *Each ℓ_i is σ-additive.*

(ii) (CE) *The family is collectively exhaustive.*

(iii) (Yosida–Hewitt) $\ell_p = 0$ *for every i.*

(iv) (Stone support) $\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1$.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 4.3. (ii) $\Leftrightarrow$ (iii) is the Yosida–Hewitt decomposition [15]. (iii) $\Leftrightarrow$ (iv) is Proposition 5.5. $\square$

When any of these holds, the Carathéodory and Stone extensions (Theorems 3.3 and 5.6) agree on $\mathcal{C}$ and hence on $\sigma(\mathcal{C})$ by Lemma 3.2.

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